Modeling Daily AQI Using Pollutant Data Across U.S.

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1. Introduction

Air quality plays a central role in public health, environmental protection, and other related decisions about the climate. In the United States, the Air Quality Index (AQI) is used to communicate the severity of air pollution and its potential health impacts. AQI is determined by measuring fine particulate matter ($PM_{2.5}$), ozone (O_3), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), and carbon monoxide (CO), the five major pollutants regulated under the Clean Air Act. Each pollutant arises from different sources and processes, and the pollutant with the highest sub-index on a given day becomes the AQI driver.

With increasingly variable atmospheric conditions, understanding air quality patterns across the United States has become more challenging. Although AQI is an established metric, its daily behavior is shaped by a range of nonlinear elements such as meteorology, regional pollution transport, wildfire smoke, and chemical interactions among pollutants. Because of this, AQI is useful for examining how data-driven models will respond to these complexities. Specifically, building a machine learning model that estimates AQI solely from pollutants assesses how much variability can be captured from these measurements alone. Identifying which pollutants have the strongest influence on AQI can also help expose factors that matter most during specific, concentrated pollution episodes.

2. Data

For this project, I used publicly available air quality datasets from the U.S. Environmental Protection Agency (EPA) AirData archive, which provides daily pollutant measurements collected from regulatory monitoring stations across the country. The data I chose spans the full year 2022 and includes measurements of the following five pollutants:

1. $PM_{2.5}$ (fine particulate matter)
   - Particles smaller than 2.5 µm capable of penetrating deep into the respiratory system.
   - Often the dominant driver of AQI during wildfire smoke events or stagnant winter conditions.
2. Ozone (O_3)
   - A secondary pollutant formed through photochemical reactions involving nitrogen oxides and volatile organic compounds.
   - Levels peak during warm, sunny periods and heavily influence summertime AQI.
3. Nitrogen Dioxide (NO_2)
   - A combustion byproduct primarily associated with vehicles, power plants, and industrial sources.
   - Serves as both a traffic-related pollutant and a precursor in ozone chemistry.
4. Carbon Monoxide (CO)
   - Produced by incomplete combustion, with the highest levels near roadways or areas

with wintertime heating emissions.
● Rarely determines AQI on a national scale but remains an informative indicator of combustion activity.

5. Sulfur Dioxide (SO_2)
   ● Emitted from power generation and industrial processes.
   ● Due to long-term emissions reductions, SO_2 concentrations have been trending lower and, in 2022, were very low in most counties.

A separate Daily AQI Summary file was used to provide the regulatory AQI value for each county and day. All pollutant datasets include station-level observations indexed by State Code, County Code, Site Number, and Date Local. These datasets were loaded using pandas and cleaned to retain only the necessary columns: measurement date, site identifiers, and the daily mean pollutant concentration. Each pollutant file was then merged with the others using shared location and date identifiers. An outer merge was used to preserve stations that reported some pollutants but not others. After aligning pollutant measurements with the AQI dataset, the data was aggregated to the overall county-per-day level, since many counties contain multiple monitoring sites. To create one unified pollutant value per county per day, measurements were averaged across all sites within the same county. This step ensured that the modeling dataset reflected county-scale conditions, which is consistent with how AQI is reported and interpreted. Rows containing missing pollutant or AQI values were removed to ensure a complete dataset for model training.

After preprocessing, the final cleaned dataset consisted of 27,987 county-day observations, the five predictor variables, and the one target variable.

3. Modeling

The goal of this analysis was to evaluate how well several commonly used regression algorithms could estimate daily AQI using routine pollutant measurements. To do this, I tested four models that spanned both linear and nonlinear approaches, with each method offering a different strength. For example, the linear models, Ridge Regression and Lasso Regression, provide interpretable, coefficient based relationships between pollutants and AQI, while the tree-based models, Random Forest and Gradient Boosting, are designed to capture more complex interactions.

Before training the models, the predictor variables ($PM_{2.5}$, O_3, NO_2, CO, and SO_2) were isolated from the cleaned dataset, and the AQI column was set as the target variable. The data was then split into training and testing sets using an 80/20 ratio. This ensured that model performance could be evaluated on samples not seen during training.

Ridge Regression is a linear model that uses L2 regularization to discourage large coefficients when predictors are correlated, a common issue in environments where pollutants co-occur. Because Ridge is sensitive to feature scaling, all pollutant variables were standardized before fitting the model. The regularization strength (α) was set to 1.0. Ridge provides a baseline for understanding how much AQI can be represented using purely linear relationships among pollutants.

Lasso is another linear method, but it uses L1 regularization, which shrinks some coefficients toward zero. In environmental datasets where certain pollutants have limited influence, like SO_2, Lasso can help identify which predictors meaningfully contribute to AQI estimation. As with Ridge, the input variables were standardized before training, but α was set to 0.01 because the two models regularize differently.

Random Forest is a method that builds decision trees using different subsets of the data and combines their predictions to better capture nonlinear relationships. Specifically, a Random Forest Regressor was used in this project, since AQI is a continuous variable. Because tree-based models do not require scaling, the raw pollutant values were used directly. The model was configured with 200 trees and a fixed random seed (42) to ensure reproducibility.

Gradient Boosting builds a sequential set of decision trees, where each tree focuses on reducing the residual errors of the previous model. Unlike Random Forest, the Gradient Boosting Regressor learns in a stage-wise fashion, enabling it to capture more structured patterns in the data when those patterns are strong and consistent. This regressor was trained using 100 estimators and the default learning rate. As with Random Forest, it operates on the raw, unscaled pollutant values.

Each model's performance was assessed using three metrics: Mean Absolute Error; Root Mean Squared Error; and R^2. Mean Absolute Error (MAE) measures the average magnitude of the prediction errors, which demonstrates how much, on average, the model's AQI estimates deviate from the true values. Root Mean Squared Error (RMSE) emphasizes larger errors, which can show how well a model handles days with high AQI. Finally, R^2 measures the proportion of total AQI variance that is explained by the pollutant predictors, giving a sense of how well the model captures the overall patterns in AQI.

4. Results

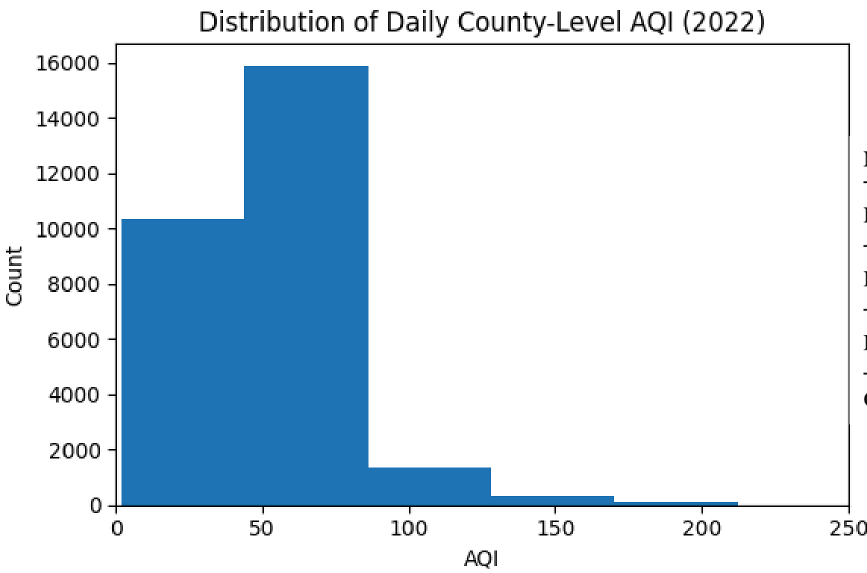

Figure 1. Histogram of daily AQI across U.S. counties in 2022

Model	MAE	RMSE	R^2
Ridge Regression	9.171	15.295	0.515
Lasso Regression	9.167	15.292	0.515
Random Forest	⭐ 7.718	⭐ 14.523	⭐ 0.562
Gradient Boosting	7.980	16.412	0.441

Figure 2: Model Performance Table

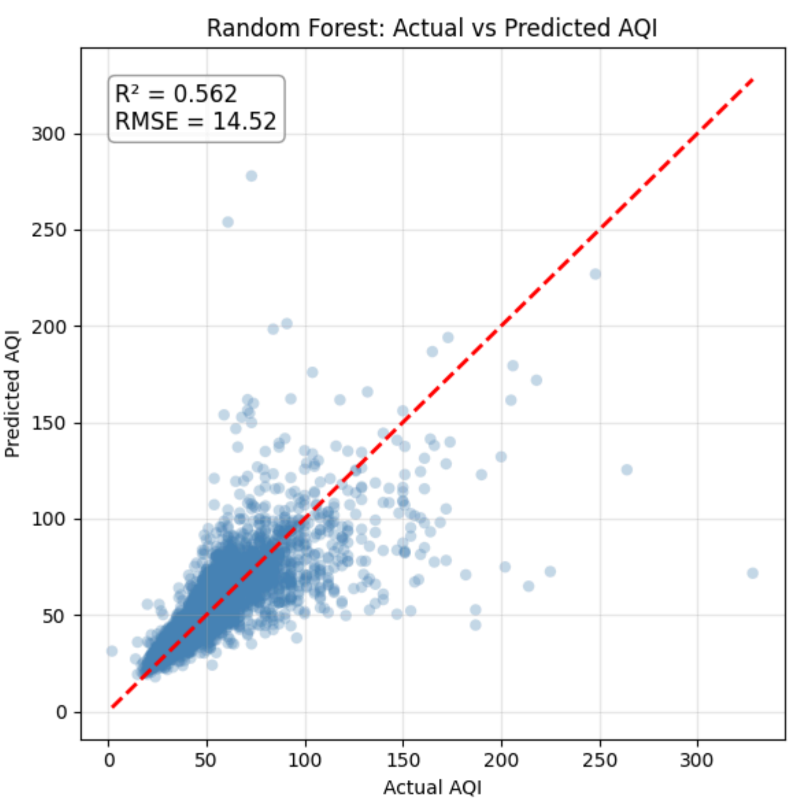

Figure 3: Scatter Plot of Predicted AQI Values vs. Observed Values (Random Forest)

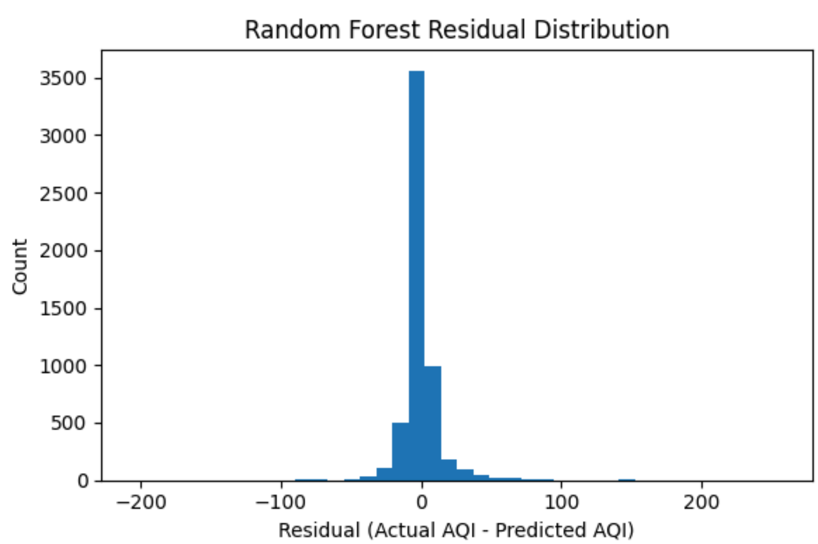

Figure 4: Histogram of Residuals

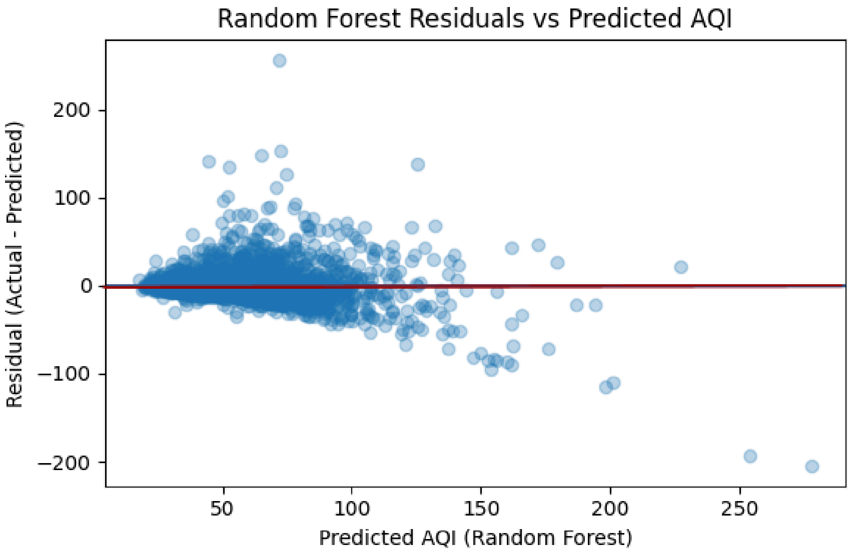

Figure 5: Residuals Plotted Against Predicted Values (Random Forest)

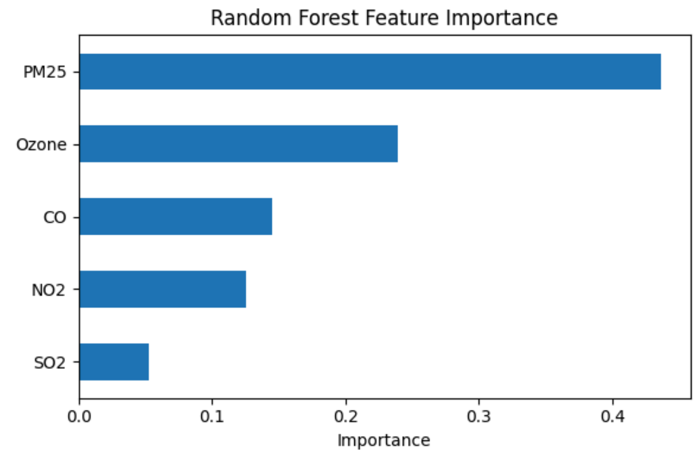

Figure 6: Bar Chart Ranking the Relative Importance of Pollutants (Random Forest)

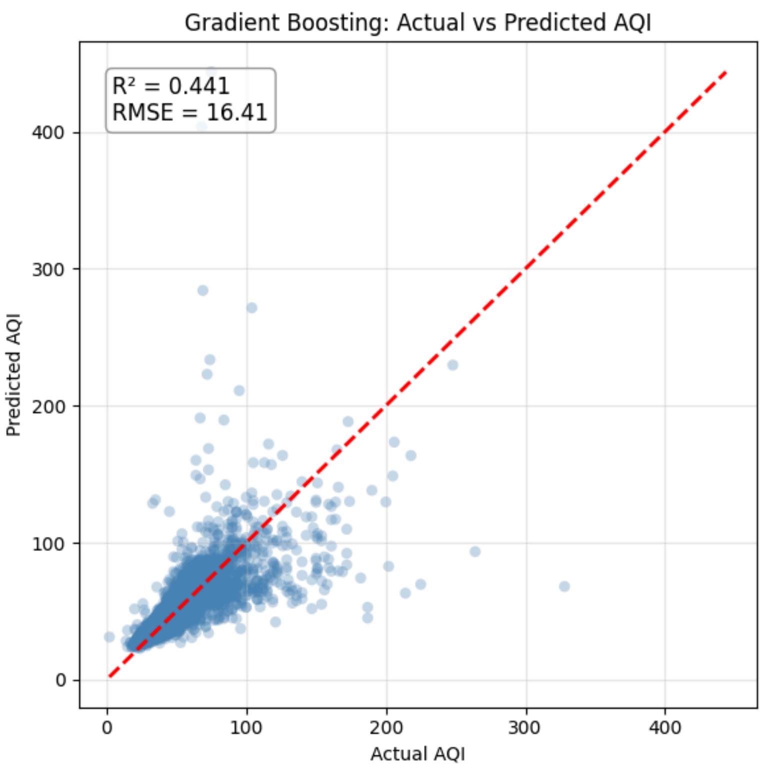

Figure 7: Scatter Plot of Predicted AQI Values vs. Observed Values (Gradient Boosting)

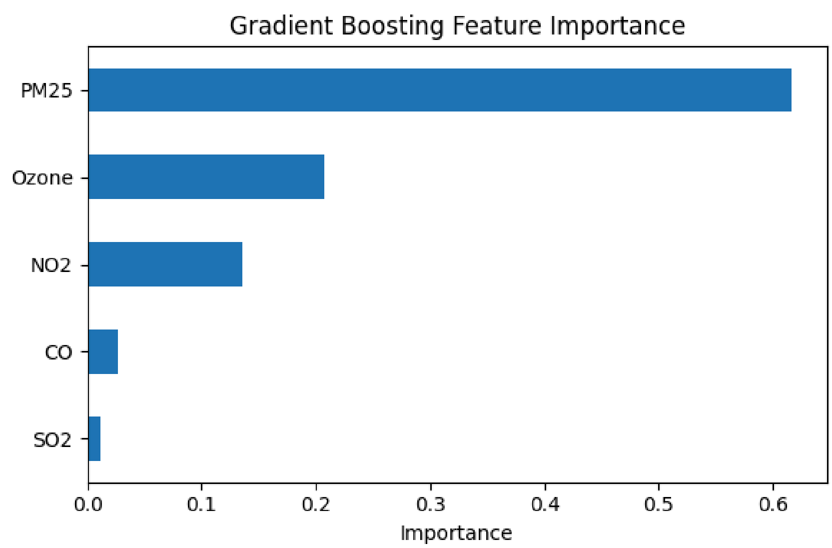

Figure 8: Bar Chart Ranking the Relative Importance of Pollutants (Gradient Boosting)

5. Discussion

Most daily AQI values observed at the county level fall in the "Good" to "Moderate" range (AQI < 100). This skewed distribution (Figure 1) helps explain why models, including the best-performing ones, tend to cluster predictions in the lower AQI ranges. From the overall model comparison in Figure 2, Random Forest had the strongest execution of the four. It produced the lowest MAE and RMSE, and the highest R^2. Ridge and Lasso performed similarly to each other, and likely captured the roughly linear portion of the pollutant to AQI relationship. However, both remained the weakest models overall because linear methods cannot capture many of the environmental factors that drive AQI. Gradient Boosting performed better than the two linear models, but not as well as Random Forest. In comparing the two, Gradient Boosting produced a lower R^2 value and larger spread in prediction errors, especially at high AQI values (Figure 7).

From the scatter plot of actual vs. predicted AQI values for Random Forest (Figure 3), the model performs well for AQI values below 100, but it begins to underestimate observations once AQI rises above 150. These higher values typically correspond to wildfire smoke or other unusual pollution episodes, which are difficult to capture just using pollutant measurements. In Figure 4, the residuals are sharply centered near zero, indicating strong performance for most county-day observations. Although the histogram does not show the full range of residuals due to implemented axis limits to improve visualization, the model's highest positive residuals occur on high AQI days, as seen in Figure 5. In addition, the widening spread of residuals signifies greater variability at higher pollution levels.

In Figure 6, the Random Forest feature importance plot shows $PM_{2.5}$ as the most influential pollutant by a substantial margin, followed by ozone. NO_2 and CO contribute moderately, while SO_2 contributes very little. $PM_{2.5}$ often drives AQI during wildfire or air stagnation conditions, whereas SO_2 levels remain extremely low across much of the United States. Figure 8 highlights that Gradient Boosting places even more weight on $PM_{2.5}$ than Random Forest does. Despite being a more flexible model than Random Forest, this emphasis likely caused Gradient Boosting to underuse the predictive information in ozone, NO_2, and CO, reducing its ability to generalize.

6. Conclusion

From this study, the following conclusions can be made:

- Random Forest performed the best overall, with the highest accuracy and R^2.
- $PM_{2.5}$ was the strongest predictor across both tree-based models.
- Ridge and Lasso performed similarly but were the weakest models.
- Gradient Boosting underperformed relative to Random Forest.
- All four models struggled to accurately predict high AQI events.

Since this dataset averages regional air quality differences, future work will benefit from building specific region or city level models to focus on local climate, geography, and emission sources. Adding variables such as weather behavior, satellite smoke indicators, or multi year data might further improve accuracy for high AQI events that pollutant data alone cannot capture. A more comprehensive system like this can help public agencies and communities better anticipate poor air quality days and respond effectively.

7. References

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